

Cover letter

Dear Editor,

We submit "EcoNiche-Opt: a locked immune-ecology transcriptomic score for multicohort prediction of immune checkpoint blockade response" for consideration as an original research article. The central premise of this study is that the main barrier to translating transcriptomic biomarkers of immune-checkpoint blockade response is not the absence of immune signatures, but the absence of an auditable validation framework that can handle heterogeneous public-cohort labels, assay platforms, endpoint definitions, and tumour-immune ecological states.

EcoNiche-Opt addresses this gap by combining manual cohort and response-label curation, endpoint-stratified evaluation, signed-rank ecological module scoring, module-interaction edges, biologically constrained optimization, leave-one-dataset-out testing, locked external validation, and a reproducible 62-gene panel-compatible scoring rule. This design makes the model refitting-free at the external-scoring stage: genes, module directions, ecological edges, coefficients, endpoint thresholds, and calibration choices are fixed before independent cohorts are scored.

In primary melanoma benchmarks, EcoNiche-Opt showed FDR-supported improvement over a predeclared eight-signature family. Locked external and panel-transfer analyses showed reproducible portability without model refitting. Single-cell localization and ecological-edge analyses linked the score to antigen presentation, T/NK effector activity, T-cell dysfunction, myeloid suppression, and stromal exclusion, supporting a coherent immune-ecology interpretation.

We believe this work will interest readers because it provides not only a predictor, but a reproducible framework for developing and validating immune-ecology biomarkers across heterogeneous ICB cohorts. The locked 62-gene qPCR/NanoString-compatible panel and open scoring implementation also create a practical route for independent computation, panel transfer, and external cohort scoring.

For transparent peer review, we opt in to publication of reviewer reports and author responses should the manuscript be accepted, in accordance with the Communications Medicine transparent peer review policy.

Sincerely,

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